

cesses to the next level. Implementing sensors and network connections on existing equipment is providing greater control and a much deeper level of insight. Additive manufacturing, or as it is more popularly known, 3D printing, is completely changing the way things like cars, electronics and even spacecrafts are manufactured. For instance, in India, WIPRO3D, which is a leading provider of metal Additive Manufacturing solutions & services, has manufactured a functional metal component using Additive Manufacturing that was used by ISRO as a part of its latest satellite which was launched into space. And unless we adopt manufacturing processes that incorporate more such cutting edge technologies that the world can get behind, it is us who'll be left behind in the manufacturing race.

R&D

When it comes to research and innovation, it has been a long standing misconception that India lags behind most western nations. India has traditionally been considered a hub for services and back-office operations for most major western corporations. In fact, the R&D and innovation hubs for these very companies, based out of U.S.A, Europe and other developed nations, have been mostly comprised of Indian researchers and scientists.

But that perception is changing, especially after the last five years. Bangalore alone houses the R&D hubs of a multitude of IT and tech companies. Apart from the sectors mentioned above like automotive and electronics, new areas like pharmaceuticals and FMCG are seeing major investments from giants like Britannia, Mondelez International and more. For instance US based Mondelez International, which manufactures products under the brand of Cadbury, plans to invest US\$ 15 million to set up a global research, development and quality hub in Thane, Maharashtra. There are many similar projects and MoUs undergoing initial setup even as this article is being drafted. You can check out a comprehensive list here: <http://dgit.in/>

randdproj. The fact that covering all of them is out of this cover story's scope stands testament to how far we've come in establishing India as a hub of innovation.

But one organisation that deserves a special mention is ISRO. From being considered nothing more than an incompetent spectator in the space race to actually establishing a cred-

(<http://dgit.in/rdstudy>), among the top 39 central universities in India, Delhi University and Banaras Hindu University have distinctly better performance among the competition (by total research papers as well as average citations per paper). But benefits of this volume of research are yet to conspicuously percolate down to the ground level in India.



In the recent PSLV launch, 29 of the 30 nanosatellites launched belonged to clients from 14 different nations

ible space programme that the world has queued to sign up for – ISRO has single handedly made a mark on the global space race with projects like the Chandrayaan and Mangalyaan. We might just also be the first country to have a privately funded space mission reach the moon, thanks to the efforts of TeamIndus.

Although we are well on our way on the road to progress, it is still true that most of these innovations are destined for, or consigned by, industries from abroad. What we need more of is research that directly affects our own country and its industries. Since most Indian universities do not figure in global rankings, comprehensive data about research at these universities is not readily available. According to a study

INDIA INC.?

In terms of development, there is no denying that there are still problems within individual areas that need to be solved – be it the lack of innovation in most startup ideas or lagging behind in cutting edge manufacturing technology. But all the information provided in this story, and the insights into the individual sectors, point towards one thing clearly – we are making progress where it matters like never before. And that trend will continue unabashed if we, as a nation, can ensure three things – adequate and up-to-date infrastructure, legislative support free from unnecessary bureaucracy and most importantly, encouragement and support for the innovative ideas and efforts among ourselves. **1**